

Operating System Types -- Guided Notes ANSWER KEY

Unit tp-3.1 | CompTIA Network+ N10-009 Obj. FC0-U71 3.1 | N10-008 FC0-U71 3.1

For instructor use / distribute after students complete the notes.

Question 1

An operating system (OS) is the software that manages a computer's _____ resources and provides services to application software. Without an OS, application programs would have to individually control every piece of _____ themselves. The OS acts as a _____ between hardware and applications.

Answer: Hardware; hardware; intermediary (abstraction layer).

Question 2

A _____ OS is designed for multi-user, multi-tasking environments on shared hardware. A _____ OS (like Windows 10/11) is designed for single-user productivity on personal computers. A _____ OS (like Android/iOS) is optimized for battery life and touchscreen interaction on handheld devices.

Answer: Server; desktop (workstation); mobile.

Question 3

Windows is developed by _____ and holds approximately _____ % of the desktop OS market. Windows uses a _____ (graphical) interface by default. The Windows Registry is a centralized database that stores _____ settings for the OS and applications.

Answer: Microsoft; ~70-75%; GUI; configuration.

Question 4

macOS is developed by _____ and runs on _____ hardware (primarily Apple Mac computers). It is built on a _____ foundation (derived from BSD Unix). This gives macOS access to a _____ terminal and Unix-compatible command-line tools.

Answer: Apple; Apple; Unix/BSD; Terminal (command-line).

Question 5

Linux is an open-source OS kernel first created by _____ in 1991. A Linux _____ is a complete OS built around the Linux kernel plus additional software. Examples include Ubuntu, _____, and _____. Linux is dominant in _____ and cloud infrastructure.

Answer: Linus Torvalds; distribution (distro); Fedora, Debian (any two); servers.

Question 6

Chrome OS is developed by _____ and is built on the _____ kernel. It is designed primarily to run _____ and Android apps. Chrome OS is popular in _____ because devices are inexpensive and easy to manage centrally via Google Admin console.

Answer: Google; Linux; web (browser); schools (K-12 education).

Question 7

An embedded OS (like _____ RTOS or VxWorks) runs inside specialized hardware with no GUI. Mobile OSes include _____ (Google, open source) and _____ (Apple, closed). The key difference between mobile and desktop OSes is that mobile OSes are optimized for _____ efficiency and _____ interfaces.

Answer: FreeRTOS; Android; iOS; power (battery); touch.

Question 8 -- Real World Application

REAL WORLD: A school district is choosing an OS for 500 new student laptops. Options are Windows 11, Chromebook (Chrome OS), and Ubuntu Linux. The laptops will be used for web browsing, Google Classroom, and basic word processing. Budget is tight. Which OS would you recommend and why? Include at least two reasons.

Answer: Chrome OS (Chromebook) is the best recommendation for this scenario. Reason 1 -- Cost: Chromebooks are significantly cheaper than Windows or MacOS devices (often \$200-350 vs \$600+). Reason 2 -- Management: Google Admin console allows one IT admin to manage all 500 devices centrally -- push policies, reset passwords, enforce content filters, and remotely wipe stolen devices. Reason 3 -- Use case alignment: the students only need a web browser and Google Classroom -- Chrome OS is built exactly for this workload. Reason 4 -- Security: Chrome OS auto-updates silently, sandboxes each app/tab, and verifies OS integrity on boot (Verified Boot), making it highly secure with near-zero IT maintenance. Ubuntu would require more technical management; Windows requires more expensive hardware and licenses.